From Avellaneda–Stoikov to GLFT: Market Making Models and Regime Switching for HIP-3 Stock Perps

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Original (Traditional Chinese)

資金費率套利聊膩了，來聊做市。

2008 年 Avellaneda 和 Stoiko 那篇論文幾乎是所有做市入門的起點。核心想法很直覺：做市商手上有庫存就有風險，所以報價不該圍著中間價，而是圍著一個「保留價格」。庫存越多、波動越大、離收盤越遠，保留價格就越往不利於你庫存的方向偏，把貨吐出去。價差則由風險厭惡和成交機率一起決定，成交機率假設隨著你離中間價的距離呈指數衰減。

AS 漂亮，但拿到幣圈直接用會出事。它有一個終止時間 T，離 T 越近價差越窄，可是加密市場 24 小時不收盤，你根本沒有 T。它也不限制庫存，理論上可以一路被單邊打穿。很多人第一次寫做市機器人就是死在這兩點。

2013 年 Gueant、Lehalle 和 Fernandez-Tapia 的 GLFT 基本上就是在補這兩個洞。它加上庫存上下限，並且推出時間趨於無限時的近似解，報價變成只跟當下庫存有關，跟時間無關。這對永續合約來說剛好，參數也變得可以實際校準：用成交資料去擬合成交強度 A 和衰減係數 k，再配上波動率和你自己的風險厭惡，就能算出每一檔庫存該報多寬、該偏多少。

可是 AS 和 GLFT 有一個共同的盲點，它們都假設中間價是隨機漫步，對價格方向一無所知。真實世界不是這樣，被聰明的單吃掉才是做市商最大的成本。所以後來 Cartea 和 Jaimungal 那一派把短期 alpha 加進來，讓報價不只依庫存偏移，也依訂單流失衡、microprice 這類預測訊號偏移。再往下走就是用 Hawkes 過程去描述成交會自我激發、一筆大單帶出一串單的現象，以及現在越來越多人在試的強化學習。

我覺得真正的機會在 HIP3 的股票永續。拿 trade.xyz 上的 SNDK 來說，美股開盤時 oracle 跟著外部價格走，休市時改成用永續自己的訂單簿算均價，等於同一個商品一週切換兩次完全不同的市場狀態。用固定參數跑 GLFT，平日校準出來的 k 和波動率到週末就失效了。

所以方向大概是這樣：平日把外部盤後價格當成 alpha 訊號來偏移報價，因為永續價格是追著它跑的；

週末外部訊號消失，成交強度變低，就該拉高風險厭惡、收窄庫存上限。

財報這種跳空事件更是要提早切換參數。誰先把 regime switching 做進做市模型裡，誰就能在這種薄盤口裡吃到別人吃不到的價差。

來源
https://hftbacktest.readthedocs.io/en/latest/
https://quant67.com/post/quant/25-market-making/25-market-making.html

English translation (X auto-translation)

Tired of talking about funding rate arbitrage? Let's talk about market making.

The 2008 paper by Avellaneda and Stoikov is practically the starting point for all market making intros. The core idea is super intuitive: market makers hold inventory, which means risk, so quotes shouldn't revolve around the mid-price, but around a "reservation price." The more inventory you have, the higher the volatility, the farther from close, the more that reservation price skews in the direction unfavorable to your inventory, to dump the goods. The spread, meanwhile, is determined by risk aversion and fill probability together, with fill probability assumed to decay exponentially with distance from the mid-price.

AS is elegant, but slapping it straight onto crypto will get you wrecked. It has a terminal time T—the closer to T, the narrower the spread—but crypto markets run 24/7 with no close, so you flat-out don't have a T. It also doesn't cap inventory, so theoretically you can get run over one-sided forever. A ton of folks' first market making bots die on exactly these two points.

The 2013 GLFT by Gueant, Lehalle, and Fernandez-Tapia basically patches those two holes. It adds inventory upper and lower bounds, and rolls out an asymptotic solution as time tends to infinity, making quotes depend only on current inventory, not time. That's perfect for perps, and the parameters become actually tunable: fit the fill intensity A and decay coefficient k from trade data, pair it with volatility and your own risk aversion, and boom—you can calculate how wide to quote and how much to skew for every inventory level.

But AS and GLFT share a common blind spot: they both assume the mid-price is a random walk, clueless about price direction. Real world ain't like that—getting picked off by smart money is the market maker's biggest cost. So later, the Cartea and Jaimungal crew started folding in short-term alpha, letting quotes skew not just on inventory but on order flow imbalance, microprice, that kind of predictive signal. Dig deeper and you get Hawkes processes to model how trades self-excite, one big order triggering a cascade, plus the reinforcement learning stuff more and more folks are experimenting with now.

I think the real opportunity is in HIP3's stock perps. Take SNDK on trade.xyz: during US stock hours, the oracle tracks external prices; after hours, it switches to averaging the perp's own order book—meaning the same product flips between two totally different market regimes twice a week. Running GLFT with fixed params? The k and vol you calibrate on weekdays crap out by the weekend.

So the direction's probably something like this: on weekdays, use post-external close prices as alpha signals to skew quotes, since perp prices chase 'em;

Weekends, external signals vanish, fill intensity drops, so crank up risk aversion and tighten the inventory cap.

Earnings gaps and other jump events? Switch params early. Whoever bakes regime switching into their MM model first gets to snag spreads in these thin books that others can't touch.

Source
https://hftbacktest.readthedocs.io/en/latest/
https://quant67.com/post/quant/25-market-making/25-market-making.html